



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 17-12-2025

Time: 09.00Am to 12.00Pm

GTM-01

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	2	2	3	3	4	4	3	5	1
6	1	7	4	8	4	9	2	10	3
11	2	12	2	13	1	14	3	15	4
16	3	17	4	18	2	19	4	20	1
21	62	22	10	23	84	24	22	25	3

PHYSICS

26	3	27	1	28	1	29	3	30	3
31	2	32	3	33	4	34	2	35	3
36	1	37	1	38	3	39	4	40	1
41	3	42	2	43	3	44	1	45	4
46	12	47	2	48	120	49	900	50	1

CHEMISTRY

51	2	52	3	53	1	54	1	55	1
56	1	57	2	58	2	59	3	60	1
61	1	62	2	63	4	64	3	65	4
66	1	67	4	68	2	69	3	70	3
71	75	72	1256	73	3	74	20	75	3



SOLUTION MATHEMATICS

1. $\frac{dy}{dx} + (2 \cot x)y = 8 \operatorname{cosec} x$

I.F. = $e^{\int 2 \cot x dx} = \sin^2 x$

$\therefore$ General solutions is

$y(\sin^2 x) = \int \sin^2 x 8 \operatorname{cosec} x dx$

$= -8 \cos x + c$

$f\left(\frac{\pi}{2}\right) = 8 \Rightarrow c = 8 \quad \therefore y = \frac{8}{1 + \cos x}$

$\therefore$ Minimum value of $y = f(x)$ is 4

2. $\sum_{i=1}^{10} (x_i - 2) = 30 \Rightarrow \sum_{i=1}^{10} x_i = 50$

$\therefore$ Mean = 5

Variance = $\frac{4}{5} \Rightarrow 25 + \frac{4}{5} = \frac{\sum x_i^2}{10}$

$\Rightarrow \sum x_i^2 = 258 \quad \sum_{i=1}^{10} (x_i - \beta)^2 = 98$

$\Rightarrow \beta^2 - 10\beta + 16 = 0 \Rightarrow (\beta - 2)(\beta - 8) = 0$

$\Rightarrow \boxed{\beta = 2} (\because \beta < 8)$

New data is $2x_i - 10 + 7\beta$

$2x_i + 4, x_i = 1, 2, \dots, 10$

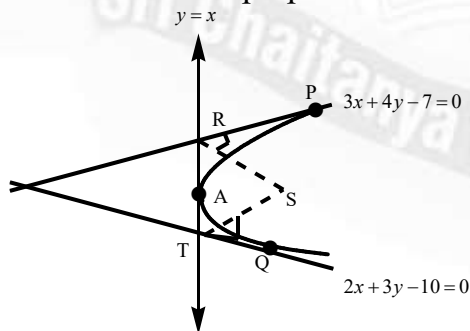
$\therefore \mu = 2(5) + 4 = 14 \quad \sigma^2 = 2^2 \left(\frac{4}{5}\right) = \frac{16}{5}$

3. $a.c_0^2 + (a+d).c_1^2 + \dots + (a+nd).c_n^2 = \frac{(2a+nd)}{2} . 2nC_n$

Here, $a = 0, d = 1, n = 100 \therefore N = \frac{0+100}{2} = 50 = 2 \times 5^2 \quad N^2 = 2^2 \times 5^4$

4. $R = (1, 1), T = (2, 2)$

Since foot of the perpendicular drawn from focus S to any tangent lies on tangent at vertex.



$\overline{SR}$ equation is $y - 1 = \frac{4}{3}(x - 1)$

$4x - 3y - 1 = 0 \dots \dots \dots (1)$



ST equation is $y - 2 = \frac{3}{2}(x - 2)$

$3x - 2y - 2 = 0 \dots\dots\dots(2)$

$\therefore S = \text{P.I. of (1) and (2)}$

$= (4, 5)$

Latus rectum $= 4a = (\text{SA})$

$= 4 \frac{|4 - 5|}{\sqrt{2}} = \frac{4}{\sqrt{2}}$

5. $\int_0^1 \left((xf(x))^2 - x\sqrt{x}f(x) \right) dx = \frac{-1}{8}$

$\Rightarrow \int_0^1 \left(\left(xf(x) - \frac{\sqrt{x}}{2} \right)^2 - \frac{x}{4} \right) dx = \frac{-1}{8}$

$\Rightarrow \int_0^1 \left(xf(x) - \frac{\sqrt{x}}{2} \right)^2 dx = 0$

$\therefore xf(x) = \frac{\sqrt{x}}{2} \quad f(x) = \frac{1}{2\sqrt{x}}$

6. $7C_4 \times D_3 + 7C_5 \times D_2 + 7C_6 \times 0 + 7C_7 \times 1$

Where D_r Derrangement of r things $= 35 \times 2 + 21 \times 1 + 0 + 1 = 92$

7. $P = (\sqrt{50} \cos \theta, 5 \sin \theta)$

$= (x_1, y_1)$

$(SP)(S'P) = (a - ex_1)(a + ex_1)$

$= a^2 - e^2 x_1^2$

$a^2 = 50, e = \frac{1}{\sqrt{2}}, x_1 = \sqrt{50} \cos \theta$

$\therefore (SP)(S'P) = 50 - \frac{50 \cos^2 \theta}{2} = 25(2 - \cos^2 \theta)$

$\therefore \alpha = 25, \beta = 50$

8. differentiating on both sides w.r.t x we get

$(x^2 - 1)e^x(x^2 + 4x + 1) = e^x(f(x) + f'(x)) \Rightarrow f(x) + f'(x) = (x^2 + 4x + 1)(x^2 - 1)$

We get non-integral roots for $x^2 + 4x + 1 = 0$

$|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{16 - 4} = 2\sqrt{3}$

9. $n(S) = 5^4 \quad E_1 - \text{Matrix symmetric}$

$E_2 - \text{Matrix Skew symmetric}$

$n(E_1) = 5 \times 5 \times 5 = 125$

$n(E_2) = 5$

$n(E_1 \cap E_2) = 1$

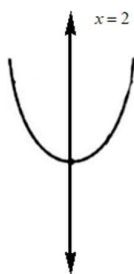
10. Z lies on a circle having $A(-\alpha), B(-\beta)$ as ends of a chord since α, β are real put $y = 0$ in

given circle $\Rightarrow x^2 + 5x + 6 = 0$

$\Rightarrow x = -2, x = -3 \Rightarrow -\alpha = -2, -\beta = -3$

$\alpha = 2, \beta = 3$

11. Graph of $y = f(x)$ is symmetric about the line $x = 2$





12. A tangent to $y^2 = 32x$

$$\Rightarrow y = mx + \frac{8}{m} \text{ which is also a tangent to } x^2 = 108y$$

$$\Rightarrow \frac{8}{m} = -(27).m^2 \Rightarrow m^3 = \frac{-8}{27} \Rightarrow m = \frac{-2}{3}$$

$$y \text{ intercept is } \frac{8}{m} = -12$$

13. $f(3x) = f(x)$

Replace x by $\frac{x}{3}, \frac{x}{3^2}, \frac{x}{3^3}, \dots$ we get

$$f(x) = f\left(\frac{x}{3}\right) = f\left(\frac{x}{3^2}\right) = \dots = f\left(\frac{x}{3^n}\right) = f(0) \text{ as } n \rightarrow \infty$$

$$\therefore f(x) = a \text{ constant function}$$

$$\therefore f(x) = 5 \forall x \in R \text{ since } f(2) = 5 \quad \therefore \operatorname{sgn}(f(x)) = 1$$

14. $\frac{(x - \sqrt{2})^2}{4} - \frac{(y - \sqrt{2})^2}{2} = 1$

Shifting the origin to $(\sqrt{2}, \sqrt{2})$

$$\text{We get } \frac{X^2}{4} - \frac{Y^2}{2} = 1$$

$$\text{Also of } \Delta ABC = \frac{1}{2}(ae - a) \times \frac{b^2}{a}$$

$$a = 2, b = \sqrt{2}, c = \sqrt{\frac{3}{2}}$$

15. $f(x + y) = f(x) + f(y)$

$$\Rightarrow f(x) = \lambda^x$$

$$f'(x) = \lambda^x \cdot \log \lambda$$

$$f'(0) = 6a \Rightarrow 1 \cdot \log \lambda = 6a$$

$$\lambda = e^{6a} \quad \therefore f(x) = e^{6ax}$$

$$\text{Now, } f''(x) - 4a f'(x) - 12f(x) = 0$$

$$\Rightarrow 36a^2 e^{6ax} - 24a^2 e^{6ax} - 12e^{6ax} = 0$$

$$\Rightarrow 36a^2 - 24a^2 - 12 = 0$$

$$\Rightarrow a = \pm 1$$

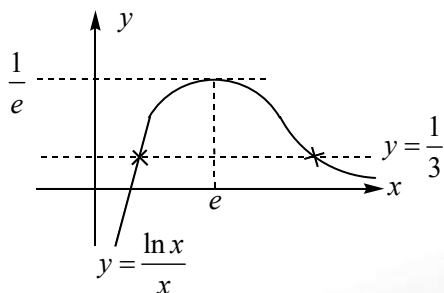
$$\Rightarrow a = -1 (\because a < 0) \quad \therefore f(x) = e^{-6x}$$

$$f(ax) = f(-x) = e^{6x}$$

$$\text{Required} = \int_0^1 e^{6x} dx = \left(\frac{e^{6x}}{6} \right)_0^1 = \frac{e^6 - 1}{6}$$

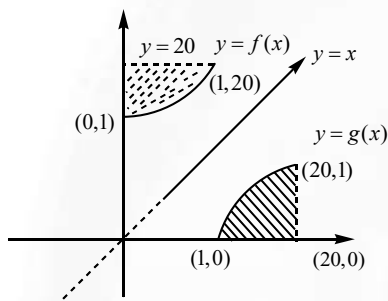
16. $x + (t \ln t - t)_0^x = \frac{x^2}{3} \Rightarrow x + x \ln x - x - (0) = \frac{x^2}{3}$

$$\Rightarrow \ln x = \frac{x}{3} \Rightarrow \frac{\ln x}{x} = \frac{1}{3}$$



17.

x_i	-1	0	1
$P(X = x_i)$	3/16	10/16	3/16

18. $f'(x) = 55x^4 + 21x^2 + 1 > 0 \forall x \in R$ 

Area of the region bounded by

 $y = f^{-1}(x)$, x -axis, $x = a$, $x = b$ is same as area of the region bounded by $y = f(x)$, y -axis, $y = a$, $y = b$

$$\text{Required area} = \int_0^1 \left(20 - (11x^5 + 7x^3 + x + 1) \right) dx = \frac{179}{12}$$

19. Let the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ bisects the circumferenceThen $(0, 0)$ lies on common chord $2gx + 2fy + c + 9 = 0 \Rightarrow c = -9$ $(1, 2)$ lies on common chord $(2g + 2)x + 2(2f + 4)y + c - 2 = 0$

$$\Rightarrow 2g + 2 + 4f + 8 - 11 = 0 \Rightarrow 2g + 4f - 1 = 0$$

$$\Rightarrow 2(-g) + 4(-f) + 1 = 0$$

 $\therefore$ locus of $(-g, -f)$ is $2x + 4y + 1 = 0$

20. $\alpha^2 + \sqrt{3}\alpha - 16 = 0$

$\gamma^2 + 3\gamma - 1 = 0$

$\alpha^{n+2} + \sqrt{3}\alpha^{n+1} - 16\alpha^n = 0$

$\gamma^{n+2} + 3\gamma^{n+1} - \gamma^n = 0$

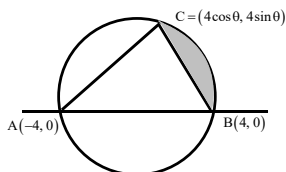
$\beta^{n+2} + \sqrt{3}\beta^{n+1} - 16\beta^n = 0$

$\delta^{n+2} + 3\delta^{n+1} - \delta^n = 0$

Adding

Adding

$$P_{n+2} + \sqrt{3}P_{n+1} - 16P_n = 0, Q_{n+2} + 3Q_{n+1} - Q_n = 0 \therefore \frac{P_{n+2} + \sqrt{3}P_{n+1}}{8P_n} = 2 \quad \frac{Q_{n+2} - Q_n}{Q_{n+1}} = -3$$

21. Let $C = (4\cos\theta, 4\sin\theta)$ 



$$\text{Area of } \Delta ABC = \frac{1}{2} \times 8 \times |4 \sin \theta| = 16 |\sin \theta| \text{ integer} \Rightarrow \sin \theta = \pm \frac{1}{16}, \pm \frac{1}{16}, \pm \frac{3}{16}, \dots, \pm \frac{15}{16}$$

$$\sin \theta = \pm 1 \quad \therefore \text{Total no of points} = 4 \times 15 + 2 = 62$$

22. Given that $x^2 = \frac{-y}{m} + x$, $m = \frac{dy}{dx}$

$$\Rightarrow \frac{dy}{dx} = \frac{-y}{x^2 - x} \Rightarrow \frac{dy}{y} = \frac{-dx}{x^2 - x} \Rightarrow \log|y| = -\int \left(\frac{-1}{x} + \frac{1}{x-1} \right) dx = \log \left| \frac{x}{x-1} \right| + c$$

Put $x = 2, y = 1 \quad \therefore |y| = \left| \frac{x}{2(x-1)} \right| \quad \therefore y = \frac{x}{2(x-1)} \quad (\because (2, 1) \in \text{curve})$

$$f(5) = \frac{5}{8} \Rightarrow f^{-1}(5/8) = 5 \quad 2f^{-1}(5/8) = 10$$

23. (No of words in which two Ms are not together) -(No of words in which two

Ms are not together and Is are together) $= \frac{4!}{2!} \times \frac{5P_2}{2!} - 3! \times \frac{4P_2}{2!} = 84$

24. Let $I = \int_0^{\pi/2} \sin^{10} x \cos^{10} x dx = 2^{-10} \int_0^{\pi/2} \sin^{10} 2x dx$

$$= 2^{-10} \cdot 2 \int_0^{\pi/4} \sin^{10} 2x dx \text{ (Queen property)} = 2^{-9} \int_0^{\pi/2} \sin^{10} t \cdot \frac{dt}{2} \quad 2x = t \quad = 2^{-10} \int_0^{\pi/2} \sin^{10} t dt$$

$$2I = 2^{-10} \int_0^{\pi/2} (\sin^{10} t + \cos^{10} t) dt \Rightarrow I = 2^{-11} \int_0^{\pi/2} (\sin^{10} x + \cos^{10} x) dx \quad \therefore n = 11 \quad 2n = 22$$

25. $(1+x)^{100} = 100C_0 + 100C_1x + 100C_2x^2 + 100C_3x^3 + 100C_4x^4 + 100C_5x^5 + 100C_6x^6 + \dots + 100C_{100}x^{100}$

Put $x = \omega$

$$(1+\omega)^{100} = 100C_0 + 100C_1\omega + 100C_2\omega^2 + 100C_3 + 100C_4\omega + 100C_5\omega^2 + 100C_6 + \dots$$

$$= (100C_0 + 100C_3 + 100C_6 + \dots)$$

$$-\frac{1}{2}(100C_1 + 100C_2 + 100C_4 + 100C_5 + \dots)$$

$$\omega = \frac{-1}{2} + \frac{i\sqrt{3}}{2}$$

$$+\frac{i\sqrt{3}}{2}(100C_1 - 100C_2 + 100C_4 - 100C_6 + \dots)$$

$$\omega^2 = \frac{-1}{2} - \frac{i\sqrt{3}}{2}$$

$$(-\omega^2)^{100} = a - \frac{b}{2} + \frac{i\sqrt{3}}{2}C$$

$$|-\omega^2|^{100} = \sqrt{\left(a - \frac{b}{2}\right)^2 + \frac{3}{4}C^2}$$

$$\therefore \left(a - \frac{b}{2}\right)^2 + \frac{3C^2}{4} = 1$$

$$\therefore \text{Ans } 3 \times 1 = 3$$

**PHYSICS**

26. Voltage drop across diode $V_D = 0.5V$;

Maximum power rating of diode $P = 100 \text{ mW} = 100 \times 10^{-3} \text{ W}$

and source voltage $V_S = 1.5V$

$$R_D = \frac{V_D^2}{P} = \frac{(0.5)^2}{100 \times 10^{-3}} = 2.5 \Omega$$

And current in diode $I_D = \frac{V_D}{R_D} = \frac{0.5}{2.5} = 0.2 \text{ A}$

Therefore, total resistance in circuit $R = \frac{V_S}{I_D} = \frac{1.5}{0.2} = 7.5 \Omega$

And the value of the series resistor = Total resistance of the circuit – Resistance of diode
 $= 7.5 - 2.5 = 5 \Omega$

27. $Y' = \overline{A.B}$ and $Y = \overline{\overline{A.B}} = A.B$

28. $\frac{4}{3}\pi r^3 \times \rho = mA \quad r \propto (A)^{1/3}$ if ρ is constant

29. $K.E. = hv - W_o \quad \text{i.e., } \frac{1}{2}mv_{\max}^2 = hv - W_o$

$$\Rightarrow \frac{1}{2}m \times (4 \times 10^6)^2 = 2hv_o - hv_o \quad \text{or, } \frac{1}{2}m \times (4 \times 10^6)^2 = hv_o$$

Another case, $2hv_o \rightarrow 5hv_o$

$$\frac{1}{2}mv_{\max}^2 = 4hv_o \Rightarrow \frac{1}{2}mv_{\max}^2 = 4 \times \frac{1}{2} \times m \times (4 \times 10^6)^2$$

$$\Rightarrow v_{\max}^2 = 64 \times 10^{12} \Rightarrow v_{\max} = 8 \times 10^6 \text{ m/s}$$

30. $b = 1$ (A universal constant)

$$\sqrt{v} = 5 \times 10^7 (41 - 1) \Rightarrow v = 4 \times 10^{18} \text{ Hz}$$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^8}{4 \times 10^{18}} = 7.5 \times 10^{-11} \text{ m}$$

31. The velocities of different colours is different in a given medium. Red and green are refracted at different angles of refraction.

32. $-\frac{1}{F} = \frac{1}{2f_1} + \frac{1}{f_m} = \frac{2}{15} + \frac{1}{\infty} \quad [\because R = \infty \text{ for plane surface, } f_m = \frac{R}{2} = \infty]$

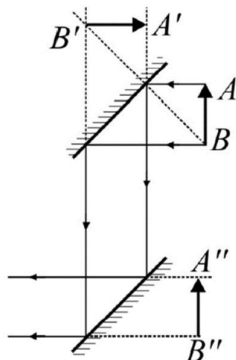
$$\therefore F = -\frac{15}{2} \text{ cm,} \quad \text{From mirror formula} \quad \frac{1}{F} = \frac{1}{u} + \frac{1}{v}$$

$$\text{i.e., } -\frac{2}{15} = \frac{1}{v} + \left(-\frac{1}{20}\right) \quad (\because u = -20 \text{ cm})$$

Solving, we get; $v = -12 \text{ cm}$

i.e., image will be formed at 12 cm to the left of AB.

33. Ray diagram of image formation is as shown in the figure below:



34. Sound waves are mechanical and light waves are electromagnetic.
 35. Angular fringe width is the ratio of fringe width to distance (D) of screen from the source,

$$\text{i.e., } \theta = \frac{\beta}{D} \Rightarrow \theta = \frac{\lambda}{d} = \frac{\left(\frac{h}{mv}\right)}{d} = \frac{h}{mvd}$$

As v is taken large, hence angular fringe width of the central maximum will decrease.

36. Due to insertion of glass plate, the image has to be shifted by 0.3 cm.

Shift produced by a glass slab of thickness t and refractive index μ is given by $t\left(1 - \frac{1}{\mu}\right)$

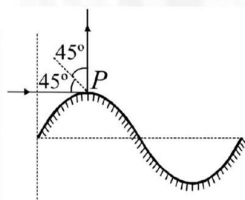
$$\text{So, } t\left(1 - \frac{1}{\mu}\right) = 0.3 \quad \text{or, } t = 0.8 \text{ cm}$$

37. Let the incidence point is $P(x, y)$

$$m = \tan 45^\circ = 1$$

$$y = 2 \sin x \quad \text{or} \quad \frac{dy}{dx} = 2 \cos x = 1 \quad \text{or} \quad \cos x = \frac{1}{2} \quad \text{or} \quad x = \frac{\pi}{3}$$

The corresponding value of $y = 2 \sin \frac{\pi}{3} = \sqrt{3}$



38. Sustained interference phenomenon takes place between two waves which have equal frequency

$$\text{Hence, } y_1 = a \sin(\omega t + \phi_1) \quad y_3 = a' \sin(\omega t + \phi_2)$$

Will give rise to sustained interference as the two waves have same frequency

39. We know that a phase change of π occurs when the reflection takes place at the boundary of denser medium. This is equivalent to a path difference of $\frac{\lambda}{2}$

Total phase difference = $\pi - \pi = 0$. i.e., two waves superimpose in phase. Hence,

$$\text{resultant amplitude} = \sqrt{I} + \sqrt{\frac{I}{4}} = \frac{3}{2} \sqrt{I}$$



So, resultant intensity $= \left(\frac{3}{2} \sqrt{I} \right)^2 = \frac{9}{4} I = \frac{9}{4} \text{ fold}$

40. Both the statements are correct according to Newton's corpuscular theory.

41. $E_0 = cB_0$

42. Here, Resistance, $R = 30 \Omega$

Inductive reactance, $X_L = 20 \Omega$ at 50 Hz

$$\therefore X_L = 2\pi\nu L \quad \therefore \frac{X_L}{\nu} = \frac{\nu'}{\nu}$$

$$X' = \frac{\nu'}{\nu} \times X_L = \left(\frac{100}{50} \right) \times 20 \Omega = 40 \Omega$$

$$\text{Impedance, } Z = \sqrt{R^2 + (X'_L)^2} = \sqrt{(30)^2 + (40)^2} = 50 \Omega$$

$$\text{Current in the coil, } I = \frac{V}{Z} = \frac{200V}{50 \Omega} = 4 A$$

43.

A $\rightarrow$ Q	At resonance, the inductive reactance (X_L) equals the capacitive reactance (X_C) so the net reactance is zero. Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$
B $\rightarrow$ R	In a purely resistive AC circuit, the voltage and current are always in phase, making the phase difference 0°
C $\rightarrow$ S	Ideal inductors and capacitors are reactive elements that store energy and release it back to the source, resulting in zero net average power dissipation over a full cycle
D $\rightarrow$ P	KCL, which states that the algebraic sum of currents entering a junction is zero, is a direct consequence of the Conservation of Electric Charge

44. Current through arm CAD, $I = \frac{V}{8}$ amp

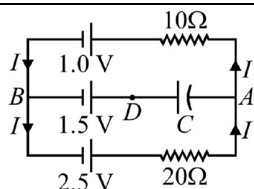
$$\text{Potential difference between C and A} = V_C - V_A = \frac{V}{8} \times 4 = \frac{V}{2} \text{ volt}$$

$$\text{Current through CBD, } I' = \frac{V}{4} \text{ amp}$$

$$\text{Potential difference between C and B} = V_C - V_B = \frac{V}{4} \times 1 = \frac{V}{4} \text{ volt}$$

$$\text{Potential difference between A and B} = V_C - V_B$$

$$\therefore V_A - V_B = V_C - V_B - (V_C - V_B) = \frac{V}{4} - \frac{V}{2} = -\frac{V}{4}$$



45.

$$20I + 10I = 2.5 - 1.0$$

$$I = \frac{1}{20} \text{ amp} \quad \therefore V_A - V_B = 2.5 - I \times 20 = 1.5 \text{ volt}$$

$$\text{But } (V_A - V_D) + (V_D - V_B) = V_A - V_B = 1.5$$

Since, there is no current through 1.5 volt cell, so,

$$V_D - V_B = 1.5 \text{ volt}$$

$$\text{Hence, } V_A - V_D = 0 \text{ volt}$$

$$46. \quad \phi = \left(\frac{q}{2\epsilon_0} - \frac{q}{6\epsilon_0} \right) \times \frac{1}{4} = \frac{q}{12\epsilon_0}$$

$$47. \quad q = \text{Area under } i-t \text{ graph} = \frac{1}{2} \times 4 \times 0.1 = 0.2 \text{ C}$$

$$\text{As } q = \frac{\Delta\phi}{R} \quad \therefore \Delta\phi = qR = (0.2 \text{ C})(10 \Omega) = 2 \text{ weber}$$

$$48. \quad \text{Given, No. of turns across primary } N_p = 50$$

$$\text{Number of turns across secondary } N_s = 1500$$

$$\text{Magnetic flux linked with primary, } \phi = \phi_o + 4t$$

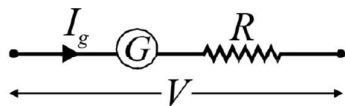
$$\therefore \text{Voltage across the primary,}$$

$$V_p = \frac{d\phi}{dt} = \frac{d}{dt}(\phi_o + 4t) = 4 \text{ volt} \quad \frac{V_s}{V_p} = \frac{N_s}{N_p} \quad \therefore V_s = \left(\frac{1500}{50} \right) \times 4 = 120 \text{ V}$$

$$49. \quad \text{Here, Resistance of galvanometer, } G = 100 \Omega$$

$$\text{Current for full scale deflection, } I_g = 30 \text{ mA} = 30 \times 10^{-3} \text{ A}$$

Range of voltmeter, $V = 30 \text{ V}$. To convert the galvanometer into an voltmeter of a given range, a resistance R is connected in series with it as shown in the figure.



$$\text{From figure, } V = I_g(G + R), \quad R = \frac{V}{I_g} - G = \frac{30}{30 \times 10^{-3}} - 100 \Omega = 1000 - 100 = 900 \Omega$$

$$50. \quad \text{Electric potential } V = -Kxy$$

$$\text{Distance of the point from origin } r = \sqrt{x^2 + y^2}$$

$$E_x = -\frac{dV}{dx} = -\frac{d}{dx}(-Kxy) = Ky \quad E_y = -\frac{dV}{dy} = -\frac{d}{dy}(-Kxy) = Kx$$

$$\therefore E_r = \sqrt{E_x^2 + E_y^2} = \sqrt{(Ky)^2 + (Kx)^2} = Kr \quad \text{i.e., } E_r \propto r$$

CHEMISTRY



51. $P_1 = Et - Br$; $P_2 = Ph - CH_2 - Br$; $P_3 = \text{>C} - Br$; $P_4 = Ph - CH_2 - I$

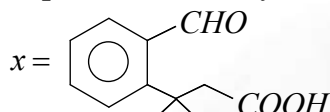
52. $CH_3 - I$ is liquid

53. Based on L.P - L.P. repulsions

54. Aldol condensation is due to presence of $\alpha - H$

55. Lysine and isoleucine are essential but only lysine has polar side chain.

56. 1st product has only one π -bond



57.

58. $Ph - NH - COCH_3$ undergoes mono bromination at para

59. Based acidic strength

60. NCERT reactions

61. I, V, VI, VII, VIII only

62. $A = [Co(en)_3]^{+2}$; $B = [Co(NH_3)_3]^{+3}$; $C = [CoCl_4]^{-2}$

63. A has no unpaired e^- s; B has 4 unpaired e^- s; C has 1 unpaired e^- ; D has 3 unpaired e^-

64. Cu^+ has d^{10} and Cu^{+2} has d^9 configuration

65. NCERT tables

66. $Gd^{+2} \rightarrow 4f^7 5d^1$

67. SO_2 is reducing and TeO_2 is oxidising

68. In 100 gm sea water,

$$\text{wt. of } CaCl_2 = 11.1 \text{ gm} \quad \Rightarrow \text{moles} = \frac{11.1}{111} = 0.1 \text{ moles} \quad i_{CaCl_2} = 3$$

$$\text{wt. of } MgCl_2 = 9.5 \text{ gm} \quad \Rightarrow \text{moles} = \frac{9.5}{95} = 0.1 \text{ moles} \quad i_{MgCl_2} = 3$$

$$\text{wt. of } NaCl = 5.85 \text{ gm} \quad \Rightarrow \text{moles} = \frac{5.85}{58.5} = 0.1 \text{ moles} \quad i_{NaCl} = 2$$

$$\text{wt. of water} = 100 - 11.1 - 9.5 - 5.85 = 73.55 \text{ gm}$$

$$\Delta T_b = K_b \times \frac{\text{moles (ions)}}{W_{\text{water}} (\text{Kg})} = \frac{0.52 \times [3 \times 0.1 + 3 \times 0.1 \times 2 \times 0.1]}{\frac{73.55}{1000}} = 5.656, \quad T_b = 105.66^\circ C$$

69. I) It is a zero order reaction as $[A_t] = [A_0] - kt$

$$K_A = \frac{[A_t] - [A_0]}{t} = \frac{0.4}{100} = 4 \times 10^{-3}$$

$$K_B = \frac{[B_t] - [B_0]}{t} = \frac{0.2}{100} = 2 \times 10^{-3}$$

$$\text{Reaction is } 2A \longrightarrow B + \text{heat} \quad K = \frac{K_A}{2} = K_B$$

II) $\therefore$ rate constant = 2×10^{-3} M/sec



III) Rate of reaction for all reactions increases with increase in temperature (both endo & exo)

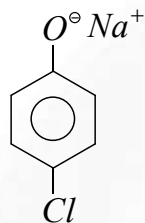
IV) All zero order reactions are complex $\therefore$ Only I & IV are true.

70. 760 torr = Total pressure at normal boiling point

$$760 = 800xX_B + 600xX_T \quad 760 = 800x + 600(1-x)$$

$$160 = 200x \quad x = \frac{160}{200} = \frac{4}{5} = 0.80$$

71. 4 is most acidic



$\therefore y$ is C_6H_4NaOCl

72. Anode : $H_2 \longrightarrow 2H^+ + 2e^-$

$$P^H = P^{K_b} + \log \frac{0.5}{0.2} = 5 + (0.7 - 0.3) = 5.4$$

$$P^H = 14 - 5.4 = 8.6$$

Cathode : $2Ag^+ + 2e^- \longrightarrow Ag$

$$\begin{aligned} \text{Cell} &= \left(E_{H_2|H^+}^0 + E_{Ag^+|Ag}^0 \right) - \frac{0.06}{2} \log \frac{[H^+]^2}{[Ag]^2} \\ &= 0.8 - 0.06 \log [H^+] + 0.06 \log [Ag^+] = 0.8 + 0.06P^H + 0.06x(-1) \\ &= 0.8 + 0.06 \times 7.6 = 0.8 + 0.456 = 1.256V = 1256 mV \end{aligned}$$

73. A, B & E

$$74. \alpha = \frac{K \times 1000}{C \Lambda^0}, \quad \alpha = \frac{4 \times 10^{-5} \times 1000}{0.004 \times 400}$$

$$\alpha = 0.025 \quad K_a \text{ is constant} \quad \therefore C_1 \alpha_1^2 = C_2 \alpha_2^2$$

$$4 \times 10^{-3} \times (0.025)^2 = 1 \times 10^{-3} (\alpha_2)^2, \quad \alpha_2 = 0.05$$

$$\begin{aligned} \pi &= iCST = 1.05 \times 10^{-3} \times 25 \times 760 \text{ (torr)} \\ &= 19.95 \sim 20 \text{ torr} \end{aligned}$$

75. Rate law : (based on rds)

$$R = K_2 [N_2O_2] [H_2]$$

$$\text{but } \frac{K_1}{K_{-1}} = K_{eqm} = \frac{[N_2O_2]}{[NO]^2},$$

$$[N_2O_2] = \frac{K_1}{K_{-1}} [NO]^2 \quad \therefore \text{Rate}$$

$$= (K_1 K_2 / K_{-1}) [NO]^2 [H_2] \Rightarrow \text{order} = 3$$